0.1 Derivatives of Polynomials

Definition 1 *Let P be a polynomial with coefficients in a field $\mathbb{K}$, defined by*

$$P = \sum_{k=0}^{n} a_k X^k.$$

1. *The derivative polynomial of P, denoted P', is defined as follows:*

$$P' = \sum_{k=1}^{n} k a_k X^{k-1}.$$

2. *For any $k \in \mathbb{N}_0$, the k-th derivative polynomial of P, denoted $P^{(k)}$, is defined as follows:*

$$\begin{cases} P^{(0)} = P \\ P^{(k)} = (P^{(k-1)})', \quad \textit{for} \quad k \geq 1. \end{cases}$$

Example 2

1. *Let $P = X^3 + 2X^2 + iX \in \mathbb{C}[X]$, then $P^{'} = 3X^2 + 4X + i$, $P'' = 6X + 4$, $P''' = 4$ and $P^{(k)} = 0$ for $k \geq 4$.*

2. *In $\mathbb{F}_p[X]$, the derivative polynomial of $P = X^p + X$ is:*

$$pX^{p-1} + 1 = 1.$$

Notice that although the degree of the polynomial P is p, the degree of the derivative is 0.

Proposition 3 *Let $P, Q \in \mathbb{K}[X]$, and $\lambda \in \mathbb{K}$, then we have the following properties:*

1. $(P + Q)' = P' + Q'$.
2. $(\lambda P)' = \lambda P'$.
3. $(PQ)' = P'Q + PQ'$.
4. $P^{(\ell)} = 0$ *if and only if* $\deg P < \ell$ $(\ell \in \mathbb{N})$.
5. $(P^\ell)' = \ell P' P^{\ell-1}$ $(\ell \in \mathbb{N})$.

Proof. Let $\lambda \in \mathbb{K}$, $\ell, m \in \mathbb{N}$, and $P = \sum_{k=0}^{n} a_k X^k, Q = \sum_{k=0}^{m} b_k X^k$ with $n \geq m$.

1. We have $P = \sum_{k=0}^{n} a_k X^k$ and $Q = \sum_{k=0}^{m} b_k X^k$, which give

$$P^{'} = \sum_{k=1}^{n} k a_k X^{k-1},\ Q^{'} = \sum_{k=1}^{m} k b_k X^{k-1}.$$

Then

$$\begin{aligned} P' + Q' &= (\sum_{k=1}^{n} k a_k X^{k-1} + \sum_{k=1}^{m} k b_k X^{k-1}) \\ &= \sum_{k=1}^{n} k(a_k + b_k) X^{k-1} = (P+Q)^{'}. \end{aligned}$$

2. We have $P = \sum_{k=0}^{n} a_k X^k$. This gives $P' = \sum_{k=1}^{n} k a_k X^{k-1}$.
Then

$$\begin{aligned} \lambda P' &= \lambda(\sum_{k=1}^{n} k a_k X^{k-1}) \\ &= (\sum_{k=1}^{n} k \lambda a_k X^{k-1}) \\ &= (\sum_{k=0}^{n} \lambda a_k X^k)' \\ &= (\lambda P)'. \end{aligned}$$

3. Let $P = \sum_{k=0}^{n} a_k X^k, Q = \sum_{k=0}^{m} b_k X^k$. Then

$$PQ = \sum_{n \geq 0} (\sum_{i=0}^{n} a_i b_{n-i}) X^n.$$

Coputing derivative, we get

$$(PQ)^{'} = \sum_{n \geq 1} n(\sum_{i=0}^{n} a_i b_{n-i}) X^{n-1}, \tag{1}$$

$$P'Q = \sum_{n \geq 1} (\sum_{i=0}^{n} i a_i b_{n-i}) X^{n-1}, \tag{2}$$

and

$$PQ' = \sum_{n \geq 1} (\sum_{i=0}^{n} (n-i) a_i b_{n-i}) X^{n-1}. \tag{3}$$

The addition of the last two equalities gives

$$P'Q + PQ' = \sum_{n\geq 1}(\sum_{i=0}^{n} ia_i b_{n-i})X^{n-1} + \sum_{n\geq 1}(\sum_{i=0}^{n}(n-i)a_i b_{n-i})X^{n-1} \quad (4)$$
$$= \sum_{n\geq 1} n(\sum_{i=0}^{n} a_i b_{n-i})X^{n-1}.$$

Then comparing (2) and (4), we get

$$(PQ)^{'} = PQ^{'} + P^{'}Q.$$

4. Let $P = \sum_{k=0}^{n} a_k X^k \in \mathbb{K}[X]$ and $\ell \in \mathbb{N}$ with $n < \ell$. We have

$$P' = (\sum_{k=0}^{n} a_k X^k)^{'} = \sum_{k=1}^{n} k a_k X^{k-1},$$
$$P^{''} = (P')' = \sum_{k=2}^{n} k(k-1)a_k X^{k-2},$$
$$\vdots$$
$$P^{(n)} = n!a_n$$
$$\vdots$$
$$P^{(\ell)} = 0 \text{ for } \ell > n.$$

Conversely, If $P^{(\ell)} = 0$. The first derivative $P'(X)$ is a polynomial of degree $n-1$. The second derivative $P''(X)$ is a polynomial of degree $n-2$. Continuing this process, after n derivatives, we obtain:

$$P^{(n)}(X) = n!a_n,$$

which is a nonzero constant (since $a_n \neq 0$). When we differentiate $P(X)$, $n+1$ times, we get:

$$P^{(n+1)}(X) = 0.$$

Then if $P^{(\ell)}(x) = 0$ for some integer ℓ, then ℓ must be greater than or equal to n.

5. By induction on n. For $n = 1$. We have $(P^1)' = 1 \times P^0 P' = P'$, which is true. Suppose that the formula is true for n and prove it for $n+1$.

$$(P^n)' = nP^{n-1}P^1.$$

$$(P^{n+1})' = (PP^n)'$$
$$= P'P^n + P(P^n)' \quad \text{(according to 3)}.$$

Replacing by the induction hypothesis we get

$$(P^{n+1})' = P'P^n + nPP^{n-1}P'$$
$$= (P^n + PnP^{n-1})P'$$
$$= (n+1)P^nP'.$$

Then the formula is true for $n+1$.

■

Proposition 4 (Leibniz formula) *For any $P, Q \in \mathbb{K}[X]$, and any $n \in \mathbb{N}$, we have*

$$(PQ)^{(n)} = \sum_{k=0}^{n} C_n^k P^{(k)} Q^{(n-k)}. \tag{5}$$

Proof. By induction on n. For $n = 1$. We have

$$(PQ)' = P'Q + PQ'$$
$$= \sum_{k=0}^{1} \binom{1}{k} P^{(k)} Q^{(1-k)}.$$

Then, according to 3) of Proposition (3) the formula is true . Assume that the formula is true for n and prove it for $n+1$.

We calculate the derivative of

$$(PQ)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} P^{(k)} Q^{(n-k)}.$$

We get

$$\begin{aligned}
(PQ)^{(n+1)} &= ((PQ)^{(n)})' \\
&= \sum_{k=0}^{n} (C_n^k P^{(k)} Q^{(n-k)})' \\
&= \sum_{k=0}^{n} C_n^k (P^{(k)} Q^{(n-k)})' \\
&= \sum_{k=0}^{n} C_n^k (P^{(k+1)} Q^{(n-k)} + P^{(k)} Q^{(n+1-k)}) \text{ (according to 3) of (3))} \\
&= \sum_{k=0}^{n+1} C_{n+1}^k P^{(k)} Q^{(n+1-k)}.
\end{aligned}$$

Then the formula is true for $n+1$. ■

Theorem 5 (Taylor formula) *Let $P \in \mathbb{K}[X]$ with $\deg P = n \geq 0$ and $\alpha \in \mathbb{K}$. Then*

$$P = \sum_{k=0}^{n} \frac{P^{(k)}(\alpha)}{k!}(X-\alpha)^k. \tag{6}$$

The formula (6) is called *Taylor expansion* of P in α.

Proof. By induction on the degree of the polynomial P. The formula is easy to establish for $\deg P = n = 0$. Assume that the formula is true for any polynomial of degree n, and show that it is true for any polynomial of degree $n+1$. Let $P \in \mathbb{K}[X]$ with $\deg P = n+1$. The Euclidean algorithm of P by $X-\alpha$ gives:

$$P = (X-\alpha)Q + P(\alpha). \tag{7}$$

Since Q is of degree n, the induction hypothesis implies that:

$$Q = \sum_{k=0}^{n} \frac{Q^{(k)}(\alpha)}{k!}(X-\alpha)^k.$$

Then

$$\begin{aligned} P &= (X-\alpha)\sum_{k=0}^{n} \frac{Q^{(k)}(\alpha)}{k!}(X-\alpha)^k + P(\alpha) \\ &= \sum_{k=0}^{n} \frac{Q^{(k)}(\alpha)}{k!}(X-\alpha)^{k+1} + P(\alpha) \end{aligned}$$

$$= P(\alpha) + \sum_{k=1}^{n+1} \frac{Q^{(k-1)}(\alpha)}{(k-1)!}(X-\alpha)^k. \tag{8}$$

Applying Leibniz's formula for (7), gives, for any $k \in \mathbb{N}$:

$$\begin{aligned} P^{(k)} &= ((X-\alpha)Q)^{(k)} \\ &= \sum_{\ell=0}^{k} C_k^{\ell}(X-\alpha)^{(\ell)}Q^{(k-\ell)} \\ &= Q^{(k)}(X-\alpha) + kQ^{(k-1)}, \end{aligned}$$

which gives for all $k \geq 1$

$$P^{(k)}(\alpha) = kQ^{(k-1)}(\alpha).$$

Then

$$Q^{(k-1)}(\alpha) = \frac{P^{(k)}(\alpha)}{k}.$$

Replacing in (8), we obtain

$$
\begin{aligned}
P &= P(\alpha) + \sum_{k=1}^{n+1} \frac{P^{(k)}(\alpha)}{k!}(X-\alpha)^k \\
&= \sum_{k=0}^{n+1} \frac{P^{(k)}(\alpha)}{k!}(X-\alpha)^k.
\end{aligned}
$$

■

Note that taking $\alpha = 0$ in Taylor formula gives the relation

$$a_k = \frac{P^{(k)}(0)}{k!}; 0 \leq k \leq n,$$

where a_k are the coefficients of P.

Example 6 *Taylor expansion of the polynomial*

$$P = 4 + X + 3X^2 + X^3$$

in $\alpha = 2$ *is*

$$P = 26 + 25(X-2) + 18(X-2)^2 + 24(X-2)^3.$$

0.2 Zeros of a Polynomial

Definition 7 *An element* $\alpha \in \mathbb{K}$ *is said to be a zero or a root of a polynomial* $P \in \mathbb{K}[X]$ *if* $\widetilde{P}(\alpha) = 0$.

Proposition 8 *Let* $P \in \mathbb{K}[X]$, *and let* $\alpha \in \mathbb{K}$. *Then,* α *is a root of* P *if and only if* P *is divisible by* $X - \alpha$.

Proof. If $(X-\alpha)$ divides P then $P = (X-\alpha)Q$, hence obviously $P(\alpha) = 0$. Conversely, assume that $P(\alpha) = 0$. Then Euclid algorithm gives $P = (X-\alpha)Q + R$ with $R = 0$ or $\deg R < \deg(X-\alpha) = 1$. Thus, R is constant. Evaluating $P(\alpha)$ gives $R = P(\alpha)$, hence $R = 0$ and $X - \alpha$ divides P. ■

Definition 9 *An element* $\alpha \in \mathbb{K}$ *is said to be a root of multiplicity* $r \in \mathbb{N}$ *of a polynomial* $P \in \mathbb{K}[X]$, *if* P *is divisible by* $(X-\alpha)^r$ *but not by* $(X-\alpha)^{r+1}$. *We say that* α *is a simple root if* $r = 1$, *a double root if* $r = 2$, *a triple root if* $r = 3$ *etc.*

Example 10 *In* $\mathbb{R}[X]$, *let* $P = -1 - 2X + 2X^3 + X^4$. *We can verify that* $\alpha = -1$ *is a root of multiplicity* 3. *We have* $(X+1)^3$ *dividing* $P = -1 - 2X + 2X^3 + X^4$. *We have* $P = (X+1)^3(X-1)$. *But* $(X+1)^4$ *does not divide* P *since* $P = 1 \times (X+1)^4 + (-2X^3 - 6X^2 - 6X - 2)$.

Proposition 11 *Let* $\alpha \in \mathbb{K}$, $r \in \mathbb{N}$, *and* $P \in \mathbb{K}[X]$. *The following assertions are equivalent.*

1. *α is a root of of multiplicity r of P.*
2. *There exists Q a polynomial with coefficients in $\mathbb{K}$ such that $P=(X-\alpha)^r Q$ and $Q(\alpha)\neq 0$.*
3. *$P(\alpha)=P^{'}(\alpha)=\cdots=P^{(r-1)}(\alpha)=0$ and $P^{(r)}(\alpha)\neq 0$.*

Proof.

1. $(1)\Rightarrow(2)$. If α is a root of multiplicity r of P, then $(X-\alpha)^r$ divides P but $(X-\alpha)^{r+1}$ does not divide P. Therefore,
$$P=(X-\alpha)^r Q$$
for some polynomial Q. Suppose $Q(\alpha)=0$. Then,
$$Q=(X-\alpha)Q_1$$
and hence,
$$P=(X-\alpha)^{r+1}Q_1,$$
which contradicts the fact that $(X-\alpha)^{r+1}$ does not divide P.

2. $(2)\Rightarrow(3)$. We have
$$P=(X-\alpha)^r Q$$
with $Q(\alpha)\neq 0$. Substituting α in the successive derivatives of P implies that α is a root of multiplicity $r-1$ of P', $r-2$ of P'', ..., $r-(r-1)=1$ of $P^{(r-1)}$, i.e., α is a simple root of $P^{(r-1)}$ and not a root of $P^{(r)}$.

3. $(3)\Rightarrow(1)$. Writing Taylor formula at α gives
$$P=\sum_{k=0}^{n}\frac{P^{(k)}(\alpha)}{k!}(X-\alpha)^k \quad (n=\deg P),$$
we can rewrite this as
$$\begin{aligned}P&=\sum_{k=r}^{n}\frac{P^{(k)}(\alpha)}{k!}(X-\alpha)^k\\&=(X-\alpha)^r\left(\frac{P^{(r)}(\alpha)}{r!}+\sum_{k=r+1}^{n}\frac{P^{(k)}(\alpha)}{k!}(X-\alpha)^{k-r}\right)\\&=(X-\alpha)^r Q.\end{aligned}$$
where
$$Q=\frac{P^{(r)}(\alpha)}{r!}+\sum_{k=r+1}^{n}\frac{P^{(k)}(\alpha)}{k!}(X-\alpha)^{k-r}.$$
Thus,
$$Q(\alpha)=\frac{P^{(r)}(\alpha)}{r!}\neq 0,$$
so $(X-\alpha)^{r+1}$ does not divide P. Therefore, α is a root of multiplicity r of P.

■

Example 12

1. *In* $\mathbb{C}[X]$, *let*
$$P(X) = 4 + 4X^2 + X^4.$$
Then
$$P(X) = (X^2 + 2)^2 = (X - i\sqrt{2})^2(X + i\sqrt{2})^2.$$
So $i\sqrt{2}$ *and* $-i\sqrt{2}$ *are roots of* P *of multiplicity* 2 *since* $P(\pm i\sqrt{2}) = P'(\pm i\sqrt{2}) = 0$ *and* $P''(\pm i\sqrt{2}) \neq 0$.

In $\mathbb{R}[X]$, let
$$P = 16 + -16X - 16X^2 + 18X^3 - 7X^4 + X^5.$$

We have
$$P(2) = P'(2) = P''(2) = P'''(2) = 0 \text{ and } P''''(2) \neq 0.$$

Thus, 2 is a root of multiplicity 4 of P, and we have
$$P = (X - 2)^4(X + 1).$$

Definition 13 *Two polynomials* P *and* Q *with coefficients in a field* $\mathbb{K}$ *are called associated if there exists* $\lambda \in \mathbb{K}^*$ *such that* $P = \lambda Q$.

Example 14 *In* $\mathbb{C}[X]$, *the polynomials* $P = 2iX^3 + 2X^2 - (i+1)X + 4$ *and* $Q = -2X^3 + 2iX^2 + (-1+i)X + 4i$ *are associates since* $Q = iP$.

Proposition 15 *Let* A *and* B *be two polynomials with coefficients in a field* $\mathbb{K}$. *Then* A *and* B *are associates if and only if* A *divides* B *and* B *divides* A.

Proof. Suppose that we have $B = AP$ and $A = BQ$ for some $P, Q \in \mathbb{K}[X]$. Then
$$B = (PQ)B.$$
If $B = 0, A = 0$ and so A and B are associated. If $B \neq 0$, $PQ = 1$ so $\deg P = \deg Q = 0$. meaning that P and Q are constants. Consequently, A and B are associates. Conversely, If A and B are associates. Then $A = \lambda B$ for some $\lambda \in \mathbb{K}^*$ which implies B divides A and since $B = \frac{1}{\lambda}A$, we also have A divides B. ■

Remark 16

1. *The relation "associate with" between polynomials is an equivalence relation in* $\mathbb{K}[X]$.

2. *Every nonzero polynomial* P *is associated with a unique monic polynomial, called the normalized polynomial associate to* P. *This polynomial is* $\frac{P}{a_d}$ *where* a_d *is the leading coefficient of* P.

Definition 17 *Let P be a non-constant polynomial with coefficients in $\mathbb{K}$. P is said to be irreducible in $\mathbb{K}[X]$ (or over $\mathbb{K}$) if it is divisible in $\mathbb{K}[X]$ only by nonzero constant polynomials and by its associates. A polynomial is said to be reducible if it is not irreducible.*

Example 18

1. *Any polynomial of degree 1 in $\mathbb{K}[X]$ is irreducible.*
2. *The polynomial $X^2 - 4$ is reducible over $\mathbb{Q}$, $\mathbb{R}$, and over $\mathbb{C}$.*
3. *The polynomial $X^2 + 1$ is reducible over $\mathbb{C}$, but irreducible over $\mathbb{R}$.*
4. *The polynomial $X^2 - 2$ is irreducible over $\mathbb{Q}$, but reducible over $\mathbb{R}$, and over $\mathbb{C}$.*

Proposition 19 *Let $P \in \mathbb{K}[X]$ with $\deg P = n \geq 0$ and $\alpha_1, \alpha_2, \ldots, \alpha_p \in \mathbb{K}$ be the distinct roots of P, with multiplicities $r_1, r_2, \ldots, r_p$. Then there exists $Q \in \mathbb{K}[X]$ such that*

$$P = (X - \alpha_1)^{r_1}(X - \alpha_2)^{r_2} \cdots (X - \alpha_p)^{r_p} Q \quad \textit{with} \quad Q(\alpha_i) \neq 0, \quad 1 \leq i \leq p.$$

Proof. Since for each $1 \leq i \leq p$, α_i is a root of multiplicity r_i, $(X - \alpha_i)^{r_i}$ divides P but $(X - \alpha_i)^{r_i+1}$ does not divide P. On the other hand, the linear polynomials $(X - \alpha_i)$, for $1 \leq i \leq p$, are irreducible. Therefore, their product

$$\prod_{i=1}^{p} (X - \alpha_i)^{r_i}$$

divides P. Thus,

$$P = \prod_{i=1}^{p} (X - \alpha_i)^{r_i} Q$$

with $Q(\alpha_i) \neq 0$. Otherwise $(X - \alpha_i)^{r_i+1}$ divides P, which is a contradiction. ■

Proposition 20 *If α is a root of multiplicity r of P, then α is a root of multiplicity $r - 1$ of P'.*

Proof. We have $P = (x - \alpha)^r Q$ with $Q(\alpha) \neq 0$. Calculating the derivative of P, we get

$$\begin{aligned} P' &= r(x - \alpha)^{r-1} Q + (X - \alpha)^r Q' \\ &= (X - \alpha)^{r-1} \underbrace{(rQ + (X - \alpha)Q')}_{Q_1}. \end{aligned}$$

Calculating $Q_1(\alpha)$ we get $Q_1(\alpha) = rQ(\alpha) \neq 0$. Then, α is a root of multiplicity $r - 1$ of P'. ■

Proposition 21 *Let*

$$P = \sum_{k=0}^{n} a_k X^k \quad (n \in \mathbb{N})$$

be a polynomial with coefficients in $\mathbb{Z}$*. If* $x = \alpha/\beta$ *is an irreducible rational root of* P*, then* α *divides* a_0 *and* β *divides* a_n*.*

Proof. Suppose that $\alpha/\beta \in \mathbb{Q}$ with $\gcd(\alpha, \beta) = 1$, is a root of P. Then

$$\textstyle\sum_{k=0}^{n} a_k \dfrac{\alpha^k}{\beta^k} = 0, \quad \text{which implies } \textstyle\sum_{k=0}^{n} a_k \alpha^k \beta^{n-k} = 0.$$

$$\text{Therefore } \begin{cases} a_0\beta^n = -\alpha \sum\limits_{k=1}^{n} a_k \alpha^{k-1}\beta^{n-k} \\ a_n\alpha^n = -\beta \sum\limits_{k=0}^{n-1} a_k \alpha^k \beta^{n-1-k}. \end{cases}$$

This gives α divides $a_0\beta^n$. Since α and β are coprime then α and β^n are also coprime, so α divides a_0. Similarly β divides $a_n\alpha^n$, as it is prime with α and therefore also with α^n, then it divides a_n. ■

Example 22 *Let* $P(X) = X^3+4X+4 \in \mathbb{Q}[X]$*. Let us find the rational roots of the polynomial* P*. Since this polynomial has degree* 3*, we need only to determine whether it has any roots in* $\mathbb{Q}$*. By the rational root test of Propsition 21, the only possible rational roots are* $\pm 1, \pm 2$*, and* ± 4*. The calculations give*

$$P(1) = 9, P(-1) = -1, P(2) = 20, P(-2) = -12, P(4) = 84, P(-4) = -76.$$

Since none of these values is zero, there are no rational roots, and the polynomial is irreducible in $\mathbb{Q}[X]$*.*

Example 23 *Let* $n \in \mathbb{N}_0$*, and let* $Q = (X-1)^{n+2} - X^{2n+1} \in \mathbb{C}[X]$*. We will show that* Q *is divisible by* $X^2 - X + 1$*. As* Q *has real coefficients and* $-\,j$*,* $-\,\overline{j}$ *are the roots of* $X^2 - X + 1$*, it is necessary and sufficient to show that* $Q(-j) = 0$*. Indeed:*

$$\begin{aligned} Q(-j) &= (-j-1)^{n+2} + (-j)^{2n+1} \\ &= j^{2(n+2)} - j^{2n+1} \\ &= j^{2n+4} - j^{2n+1} \\ &= j^{2n+1} - j^{2n+1} \\ &= 0. \end{aligned}$$

0.3 Factorization of Polynomials in $\mathbb{K}[X]$

Proposition 24 *Any polynomial P of degree n in $\mathbb{K}[X]$ admits at most n roots (counted with their multiplicities).*

Proof. Let $P \in \mathbb{K}[X]$. Suppose that $\alpha_1, \alpha_2, ..., \alpha_\ell$ are the distinct roots with multiplicities $m_1, m_2, ..., m_\ell$ of P. Then the polynomials

$$Q_i = (X - \alpha_i)^{m_i} \quad (1 \leq i \leq \ell)$$

are pairwise coprime, and for any $1 \leq i \leq \ell$, the polynomial Q_i divides P. So the product

$$Q = \Pi_{i=1}^{\ell} Q_i = \Pi_{i=1}^{\ell} (X - \alpha_i)^{m_i}$$

divides P. Now, the polynomial Q has degree

$$\sum_{i=1}^{i=\ell} m_i,$$

so

$$\sum_{i=1}^{i=\ell} m_i \leq \deg P = n.$$

■

Theorem 25 (Algebra fundamental theorem) *Every non-constant polynomial in $\mathbb{C}[X]$ has at least one root in $\mathbb{C}$.*

For the proof of this very important theorem, see the excellent book by J. Dixmier [**?**].

Definition 26 *A field $\mathbb{K}$ is said to be algebraically closed if every non-constant polynomial over $\mathbb{K}$ has a root in $\mathbb{K}$.*

Proposition 27 *The field of complex numbers $\mathbb{C}$ is algebraically closed.*

Proof. This is a direct consequence of the algebra fundamental theorem. ■

Theorem 28 *Let $P \in \mathbb{K}[X]$ be a non-constant polynomial. Then there exist $\lambda \in \mathbb{K}$, monic irreducible polynomials $P_1, P_2, ..., P_n \in \mathbb{K}[X]$, and $\ell_1, \ell_2, ..., \ell_n \in \mathbb{N}^*$ such that*

$$P = \lambda P_1^{\ell_1} P_2^{\ell_2} \cdots P_n^{\ell_n}.$$

Moreover, this factorization is unique up the order of the factors.

Corollary 29 *In $\mathbb{C}[X]$, any polynomial of degree $n \geq 1$ has n roots in $\mathbb{C}$, each root counted a number of times equal to its multiplicity.*

Proof. This follows by induction on the degree n of P. ■

Corollary 30

1. *The only irreducible polynomials in* $\mathbb{C}[X]$ *are polynomials of degree* 1.
2. *The only irreducible polynomials in* $\mathbb{R}[X]$ *are polynomials of degree* 1 *and trinomials* $X^2 + aX + b$ *with* $a^2 - 4b < 0$*. The quantity* $a^2 - 4b$ *is called the discriminant of the trinomial* $X^2 + aX + b$*.*

Proof.

1. The only monic irreducible polynomials on $\mathbb{C}$ are $X - \alpha$, with $\alpha \in \mathbb{C}$. This is clear because every non-constant polynomial has a factor of this type according to the algebra fundamental theorem.
2. Due to the degree, a polynomial of the form $X - \alpha$ ($\alpha \in \mathbb{R}$) is irreducible, since its only divisors are the nonzero constant polynomials and its associates. Let $P \in \mathbb{R}[X]$ of degree greater than or equal to 2. Then P admits at least one root α in $\mathbb{C}$. Since the coefficients of P are real, $\overline{\alpha}$ is also a root, and so P is divisible by the product

$$(X - \alpha)(X - \overline{\alpha}) = X^2 - 2\,\mathrm{Re}(\alpha)X + |\alpha|^2 \in \mathbb{R}[X].$$

This conclude that the only irreducible polynomials in $\mathbb{R}[X]$ are the polynomials of degree 1 and trinomials aX^2+bX+c with discriminant $\triangle = b^2-4ac < 0$.
■

Example 31

1. *The polynomial* $P = X^3 + 2$ *is irreducible over* $\mathbb{Q}$*. Indeed, if* P *were reducible over* $\mathbb{Q}$*, it would be divisible by a polynomial of degree* 1 *in* $\mathbb{Q}[X]$*, which implies that it has a root in* $\mathbb{Q}$*. By Proposition (21), the only possible rational roots are* ± 1*. Checking these, neither* 1 *nor* -1 *are roots, so the polynomial is irreducible over* $\mathbb{Q}$*.*
2. *The polynomial* $P = X^3 + X + 1$ *is irreducible in* $\mathbb{F}_5[X]$*. Indeed, we have*

$$P(\overline{0}) = \overline{1}, \quad P(\overline{1}) = \overline{3}, \quad P(\overline{2}) = \overline{1}, \quad P(\overline{3}) = \overline{1}, \quad P(\overline{4}) = \overline{4}.$$

Since none of these values is 0, P *has no roots in* $\mathbb{F}_5$*. Thus, since* P *has degree* 3*, it is irreducible in* $\mathbb{F}_5[X]$*.*

3. *The polynomial* $P = X^3 + X^2 + 1$ *is irreducible in* $\mathbb{F}_2[X]$*. Indeed, we have*

$$P(\overline{0}) = P(\overline{1}) = \overline{1} \neq \overline{0}.$$

Since P *has no roots, and* P *has degree 3, it is irreducible in* $\mathbb{F}_2[X]$*.*

Theorem 32

1. *Any non-constant polynomial P in $\mathbb{C}[X]$ is uniquely factorized as*

$$P = \lambda \prod_{k=1}^{p} (X - \alpha_k)^{r_k},$$

where $\alpha_1, \alpha_2, \dots, \alpha_p$ are the distinct roots of P in $\mathbb{C}$, $r_1, r_2, \dots, r_p$ are their respective multiplicities, λ is its leading coefficient, with $\sum_{k=1}^{p} r_k = \deg P$,

2. *Any non-constant polynomial P in $\mathbb{R}[X]$ is uniquely factorized as*

$$P = \lambda \prod_{k=1}^{p} (X - \alpha_k)^{r_k} \prod_{\ell=1}^{q} (X^2 + a_\ell X + b_\ell)^{s_\ell},$$

where $\alpha_1, \alpha_2, \dots, \alpha_p$ are the distinct roots of P in $\mathbb{R}$, $r_1, r_2, \dots, r_p$ their respective multiplicities, λ is the leading coefficient of P, a_ℓ and b_ℓ are in $\mathbb{R}$ such that $a_\ell^2 - 4b_\ell < 0$, $s_\ell \in \mathbb{N}$ $(1 \leqslant \ell \leqslant q)$, with $\sum_{k=1}^{p} r_k + 2\sum_{l=1}^{q} s_\ell = \deg P$.

Proof.

1. Let $P \in \mathbb{C}[X]$. We prove this by induction on the degree of P. Suppose $\deg P = n$. The proposition is obvious for $n = 1$. Assume that it is true for all polynomials of degree less than or equal to n, and show that it is true for all polynomials of degree $n+1$. Let $P \in \mathbb{C}[X]$ with deg $P = n+1$. P has at least one root $\alpha \in \mathbb{C}$ and can be written as

$$P = (X - \alpha)^r Q, \quad \deg Q = n - r, \quad Q(\alpha) \neq 0.$$

r being the multiplicity of α in P. The induction hypothesis implies that Q factorizes as

$$Q = \lambda \prod_{k=1}^{p} (X - \alpha_k)^{r_k}, \quad \sum_{k=1}^{p} r_k = n - r.$$

Thus,

$$P = \lambda (X - \alpha)^r \prod_{k=1}^{p} (X - \alpha_k)^{r_k}.$$

2. Let P be a non-constant polynomial of $\mathbb{R}[X]$. Suppose $\alpha_1, \alpha_2, \dots, \alpha_p$ are the distinct real roots of P with multiplicities $r_1, r_2, \dots, r_p$ and λ is the leading coefficient of P. Then

$$P = \lambda \prod_{k=1}^{p} (X - \alpha_k)^{r_k} Q,$$

where $Q \in \mathbb{R}[X]$ and $Q(\alpha_k) \neq 0$ for $1 \leq k \leq p$. The polynomial Q has coefficients in $\mathbb{R}$ and does not admit real roots. Then whenever P

admits a root β in $\mathbb{C}\backslash\mathbb{R}$, it admits its conjugate $\overline{\beta}$ as a root with the same multiplicity. Suppose $\beta_1, \beta_2, \ldots, \beta_q$ be the distinct roots of P in $\mathbb{C}\setminus\mathbb{R}$ and $s_1, s_2, \ldots, s_q$ their respective multiplicities. Then in $\mathbb{C}[X]$ we have

$$\begin{aligned} P &= \lambda \prod_{k=1}^{p}(X-\alpha_k)^{r_k} \prod_{\ell=1}^{q}(X-\beta_\ell)^{s_\ell}(X-\overline{\beta}_\ell)^{s_\ell} \\ &= \lambda \prod_{k=1}^{p}(X-\alpha_k)^{r_k} \prod_{\ell=1}^{q}(X^2-(\beta_\ell+\overline{\beta_\ell})X+\beta_\ell\overline{\beta_\ell})^{s_\ell}. \end{aligned}$$

We put $a_\ell = (\beta_\ell + \overline{\beta_\ell})$ and $b_\ell = \beta_\ell\overline{\beta_\ell}$ for $1 \leqslant \ell \leqslant q$. Then we have $a_\ell^2 - 4b_\ell < 0$, for all $1 \leqslant \ell \leqslant q$. Thus, we obtain the following decomposition of P in $\mathbb{R}[X]$

$$P = \lambda \prod_{k=1}^{p}(X-\alpha_k)^{r_k} \prod_{\ell=1}^{q}(X^2 - a_\ell X + b_\ell)^{s_\ell},$$

where as seen above, $\alpha_1, \alpha_2, \ldots, \alpha_p$ are the distinct roots of P in $\mathbb{R}$, $r_1, r_2, \ldots, r_p$ their respective multiplicities, and λ is the leading coefficient of P, a_ℓ and b_ℓ are in $\mathbb{R}$ verifying $a_\ell^2 - 4b_\ell < 0$, $r_k, s_\ell \in \mathbb{N}_0$ for all $1 \leqslant k \leqslant p$ and $1 \leqslant \ell \leqslant q$.

■

Example 33

1. *Decompose the polynomial* $P = X^8 - 1$ *into irreducible factors in* $\mathbb{R}[X]$. *We first decompose* P *in* $\mathbb{C}[X]$ (*using eighth roots of unity*), *we have:*

$$\begin{aligned} P &= (X-1)(X+1)(X-i)(X+i)(X-\frac{\sqrt{2}-i\sqrt{2}}{2}) \\ &\times (X-\frac{\sqrt{2}+i\sqrt{2}}{2})(X+\frac{-\sqrt{2}+i\sqrt{2}}{2})(X+\frac{-\sqrt{2}-i\sqrt{2}}{2}). \end{aligned}$$

   Then grouping the conjugate factors we obtain

$$P = (X-1)(X+1)(X^2+1)(X^2+\sqrt{2}X+1)(X^2-\sqrt{2}X+1).$$

2. *Factorize* $P = X^3 + 1$ *in* $\mathbb{R}[X]$. *We have*

$$P = (X+1)(X^2-X+1).$$

3. *Decompose* $P = X^6 - 1$ *in* $\mathbb{R}[X]$. *Using the sixth roots of unity, we have*

$$P = (X-1)(X+1)(X-j)(X-j^2)(X+j)(X+j^2),$$

   where j *is the complex number satisfying* $j^3 = 1$ *with positive imaginary part. Therefore*

$$P = (X-1)(X+1)(X^2+X+1)(X^2-X+1).$$

4. *Factorize* $P = X^n - 1, \quad n \in \mathbb{N}$ *over* $\mathbb{C}$ *and* $\mathbb{R}$. *The roots of* P *in* $\mathbb{C}$ *are the* n*-th roots of unity:*

$$e^{i\frac{2k\pi}{n}} \ (0 \le k \le n-1).$$

Over $\mathbb{C}$, *we have*

$$P = \prod_{k=0}^{n-1} (X - e^{i\frac{2k\pi}{n}}).$$

Over $\mathbb{R}$, *two cases must be distinguished.*

- *If* $n = 2p$, *we get*

$$P = (X-1)(X+1) \prod_{k=1}^{p-1} \left(X^2 - 2 \left(\cos \frac{k\pi}{p} \right) X + 1 \right).$$

- *If* $n = 2p+1$, *we get*

$$P = (X-1) \prod_{k=1}^{p} \left(X^2 - 2 \left(\cos \frac{2k\pi}{2p+1} \right) X + 1 \right).$$